

MATLAB: Session 6

Vincenzo Antonio Isoldi
Francesco Giuseppe De Vitis

6 November 2025

Exercises

Part A – Linear Applications and Change of Basis

In the following, $\mathcal{E}_n = \{e_1, \dots, e_n\}$ denotes the canonical basis for $\mathbb{R}^n$.

A.1 Let f be the endomorphism of $\mathbb{R}^3$, represent by the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 1 \\ 3 & -2 & 1 \end{bmatrix},$$

on the basis $\mathcal{B} = \{(1, 1, 1), (0, 1, 1), (0, -1, 1)\}$.

A.1.1 Assign the matrix A to the variable `A`.

A.1.2 Compute $M_{\mathcal{E}_3, \mathcal{E}_3}(f)$. Assign it to the variable `Mf_E3`

A.1.3 Compute the dimension of $\ker f$ and assign it to the variable `dimkerf`.

A.1.4 Compute the dimension of $\operatorname{Im} f$ and assign it to the variable `dimimf`.

A.1.5 Compute a basis for $\ker f$ and assign it to the variable `basiskerf`.

A.1.6 Compute a basis for $\operatorname{Im} f$ and assign it to the variable `basisimf`.

A.1.7 Is f injective? Is f surjective?

A.2 Let $g : \mathbb{R}_2[x] \rightarrow \mathbb{R}_1[x]$ be the linear map defined by

$$g(1+x) = 1, \quad g(1+x^2) = 1+x, \quad f(x+x^2) = x$$

A.2.1 Let $\mathcal{B}_1 = \{1+x+x^2, 1+x^2, x+x^2\}$ be a basis of $\mathbb{R}_2[x]$ and $\mathcal{B}_2 = \{1+x, 1-x\}$ be a basis of $\mathbb{R}_1[x]$. Compute the matrix $M_{\mathcal{B}_1, \mathcal{B}_2}(g)$ and assign it to the variable `Mg_B1B2`.

A.2.2 Let $\mathcal{C}_1 = \{1, x, x^2\}$ and $\mathcal{C}_2 = \{1, x\}$ be the canonical bases of $\mathbb{R}_2[x]$ and $\mathbb{R}_1[x]$, respectively. Compute (explicitly, i.e. without using the transition matrices) the matrix $M_{\mathcal{C}_1, \mathcal{C}_2}(g)$ and assign it to the variable **Mg_C1C2**.

A.2.3 Verify that the change of basis formula holds. Namely, compute the transition matrices $M_{\mathcal{C}_1, \mathcal{B}_1}(id)$ (i.e. the transition matrix from the basis $\mathcal{C}_1$ to the basis $\mathcal{B}_1$) and $M_{\mathcal{B}_2, \mathcal{C}_2}(id)$ (i.e. the transition matrix from the basis $\mathcal{B}_2$ to the basis $\mathcal{C}_2$) and check that

$$M_{\mathcal{C}_1, \mathcal{C}_2}(g) = M_{\mathcal{B}_2, \mathcal{C}_2}(id) \cdot M_{\mathcal{B}_1, \mathcal{B}_2}(g) \cdot M_{\mathcal{C}_1, \mathcal{B}_1}(id).$$

A.2.4 Compute the dimension of $\ker g$ and assign it to the variable **dimkerg**.

A.2.5 Compute the dimension of $\text{Im} g$ and assign it to the variable **dimimg**.

A.2.6 Compute a basis for $\ker g$ and assign it to the variable **basiskerg**.

A.2.7 Compute a basis for $\text{Im} g$ and assign it to the variable **basisimg**.

A.2.8 Is g injective? Is g surjective?

A.3 Let $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the endomorphism represented, with respect to the canonical basis $\mathcal{E}_3$, by the following matrix

$$C = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

A.3.1 Assign the matrix C to the variable **C**.

A.3.2 Check that the vectors

$$\begin{aligned} v_1 &= e_2 + e_3, \\ v_2 &= e_1 + e_3, \\ v_3 &= e_1, \end{aligned}$$

form a basis for $\mathbb{R}^3$.

A.3.3 Let $\mathcal{D}$ be the basis of the previous point. Compute $M_{\mathcal{D}, \mathcal{D}}(h)$. Assign it to the variable **Mh_D**.

A.3.4 Compute the dimension of $\ker h$ and assign it to the variable **dimkerh**.

A.3.5 Compute the dimension of $\text{Im} h$ and assign it to the variable **dimimh**.

A.3.6 Compute a basis for $\ker h$ and assign it to the variable **basiskerh**.

A.3.7 Compute a basis for $\text{Im} h$ and assign it to the variable **basisimh**.

A.3.8 Is h injective? Is h surjective?

Part B – Determinant

B.1 Consider the matrix

$$M = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 1 & 4 & 0 \end{bmatrix}.$$

Assign it to the variable `M`.

B.2 Compute, using Laplace's rule, the determinant of M . Assign it to the variable `detM` and check your result using the MATLAB function `det()`.

B.3 Define the matrix N in the following way: take M , add to its second row two times the first row, then subtract from the third row the sum of the first two rows. Assign it to the variable `N`.

B.4 Compute $\det(N)$. Assign it to `detN` and compare it with `detM`. What do you notice?

B.5 Define the matrix L following the steps in **B.3**, but working with columns instead of rows. Assign it to the variable `L`.

B.6 Compute $\det(L)$. Assign it to `detL` and compare it with `detM`. What do you notice?

B.7 Define the matrix O from M , dividing the first row by 17. Assign it to the variable `O`.

B.8 Compute $\det(O)$. Assign it to `detO` and compare it with `detM`. What do you notice?

B.9 Consider the matrix P , obtained from M by swapping the first two rows. Assign it to the variable `P`.

B.10 Compute $\det(P)$. Assign it to `detP` and compare it with `detM`. What do you notice?